#	Learning Type	Justification	Input	Output
1	Supervised (Classification)	Labeled historical data exists — past customers who canceled vs stayed. Model learns from these labels.	Customer data: charges, tenure, complaints, usage, call minutes	Will cancel or not
2	Reinforcement	Self driving car tries to do itself, if it made, penalty; if not, reward, it gets better over time	Road state, location of cars, speed, traffic lights status	Actions like turn right, turn left speed up, stop
3	Supervised (classification)	Labeled historical data exists-past customers who got recommendation vs don't take recommend. Model learns from these labels	Customer data: your and others data	Recommendation lists: phone, headphone, powerbank
4	Unsupervised	No labelling data, just learns behavior, what app they are spending their time	Clock, app usage, traffic	Night users, social media users, video stream users
5	Supervised (classification)	There are lebelling data, email exists	Email text, subject, sender address, whether exists link, word pattern	Spam or not
6	reinforcement	If robot make a mistake, will get a penalty, else gets reward. No labeled examples — it learns from experience.	Position of ore pieces, sensor data, visual input, weight	Action: pick up or leave, sort out to A or B category
7	Supervised (regression)	There are data: demand signals and user response	User behavior, average monthly salary, will check whether they have credits	Best price: 45 000 sums
8	Supervised callsification	Data exists: past patients history	Past patients data, how many of them had diabetes, signs	Will have diabetes or not
9	Unsupervised clustering	Data exists, but AI finds itself what groups it should divide	User behavior features: how many tik-tokers, you tubers	Group 1: night user Group 2: social media users

10	Reinforcement	No data exists, just learns experience	Change tactics, write where he made a mistake, board state	Each action: move e4, move g9
11	Unsupervised (anomaly detection)	Fraud-not fraud may not be pre-labeled in all cases	User transaction history, call/sms patterns, internet usage	Normal, suspicious, fraud flag
12	Unsupervised (clustering)	This article was not labeled as politics or sports	News article text, headline, keywords	Topic 1: politics Topic 2 : sports
13	Supervised regression	There are past data that how much this kind of house were cost	Square footage, district, floor number, building type	House price: $99 000
14	reinforcement	Exploring time based penalties	Where he is now, what ways are open, road length	Go ahead, stop, turn left
15	Supervised classification	There are labels	Uzbek sentences, grammatics	I have done the task
16	Unsupervised (clustering)	There are no labels, model finds itself	User listening history, skip, like behaviours	Music A, Music B
17	Unsupervised (anomaly detection)	There are no “normal vs anomaly” labels	Number of requests, bandwidth usage	Normal, suspicious, anomaly alert

1. Step 2. Spotify “Discover weekly” –unclear part is there are explicit labels like user will like this song. There is no direct labeling saying.
2. Uztelecom network anomaly detection:
   anomaly is explicitly defined (supervised) or discovered statistically (unsupervised)

3. Step 3. System must detect unknown anomalies, new attack patterns constantly appear
4. Step 1. #9;11;16;17;13 were hard examples